Case Study: Optimizing Electric Traction Drives with Neural Networks

Client: Industrial Manufacturers & Automotive Engineering Firms

Prepared by: FEAAM GmbH

Subject: Embedded Benchmarking of Neural Network vs. Lookup Table-Based Current Reference Generation

Executive Summary

Modern electric traction applications, such as electric vehicles, require high-performance motor control that is both precise and hardware efficient. FEAAM GmbH has demonstrated a breakthrough by replacing traditional, memory-heavy Lookup Tables (LUTs) with a lightweight Neural Network (NN) for current reference generation.

This innovation achieves a 97.2% reduction in memory usage while providing significantly higher accuracy than traditional methods. FEAAM helps by implementing scalable AI control logic that eliminates the need for expensive memory upgrades, allowing manufacturers to deliver superior motor efficiency and extended range within existing hardware constraints.

The Challenge: The "Memory Wall" in Embedded Control

Standard Field-Oriented Control (FOC) relies on "cheat sheets" called Lookup Tables to manage motor currents. However, as motors become more complex, manufacturers face two expensive hurdles:

1. Exponential Memory Demand: Adding new variables (like temperature or aging) causes LUT size to explode, often exceeding the capacity of standard microcontrollers.
2. Accuracy Trade-offs: To save space, LUT resolution is often reduced, leading to "gaps" in data that cause efficiency losses and poor control in non-linear operating regions.

The FEAAM Solution: Neural Network Innovation

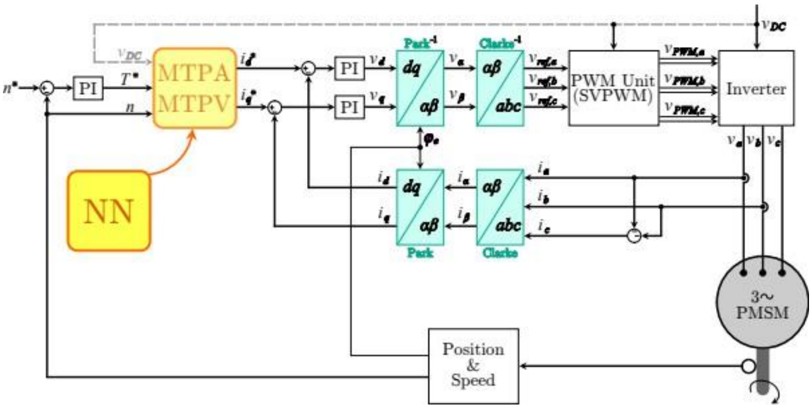


Fig. 1: Field-Oriented Control structure with NN replacing conventional MTPA/MTPV block

Instead of using massive static tables, FEAAM engineers developed a compact, manually implemented Neural Network that "learns" the optimal motor settings.

How it works:

Think of the motor's settings like a complex, 3D landscape of power.

- A standard LUT is like a grainy photograph of that landscape; the more detail you want, the bigger the photo must be.
- Our Neural Network is like a mathematical formula that can perfectly redraw that entire landscape from scratch using very little information. It "steers" the motor to its most efficient state in real-time without needing a massive database.

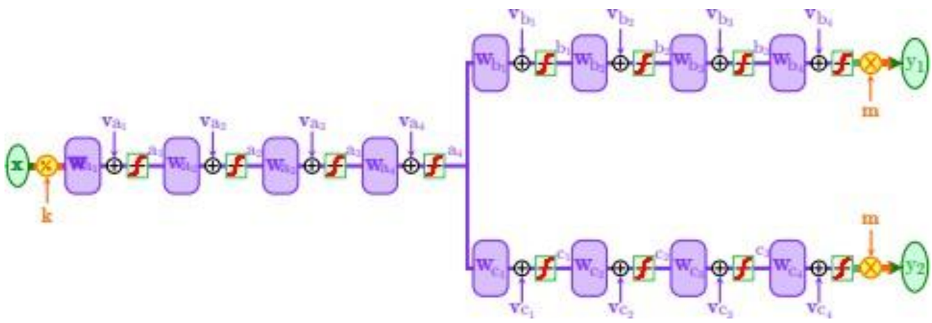
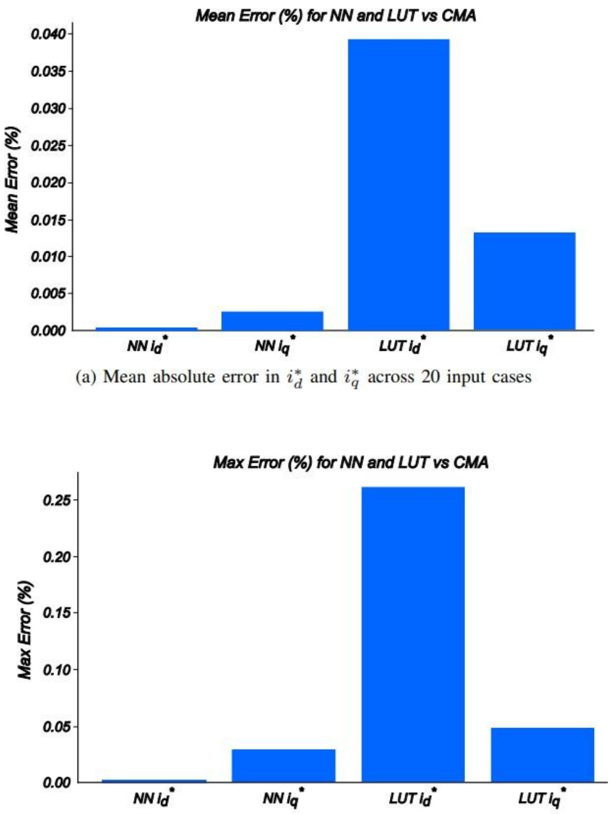


Fig. 2: NN architecture used for current reference generation

Measurable Results:

We compared our Neural Network implementation against a standard 3D-LUT on an STM32H7B3I-DK microcontroller.

Feature	Standard LUT	FEAAM Improved NN	Benefit
Flash Memory Usage	~2050 kB	~56 kB	97.2% Reduction
Control Accuracy	5.8% – 7.5% Error	< 0.06% Error	Smoother/Efficient Drive
Inference Time	0.028 ms	0.5 ms	Real-Time Compliant
Hardware Requirement	High (2 MB+)	Low (Tiny Footprint)	Lower BOM Costs



(a) Mean absolute error in i_d^* and i_q^* across 20 input cases

(b) Maximum absolute error in i_d^* and i_q^* across 20 input cases

Fig. 3: Comparison of statistical accuracy metrics (mean and maximum absolute error) for NN and LUT implementations with respect to the CMA reference over 20 test inputs

The Bottom Line for Your Business

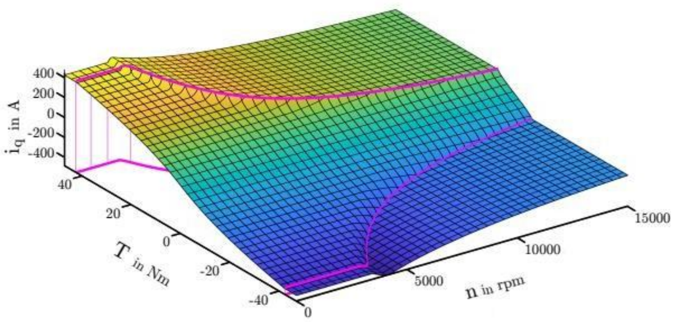


Fig. 4 Optimal current reference surface for i_q^* computed via constrained minimization

This technology provides a massive competitive advantage for next-generation electric drives. By choosing FEAAM's design approach, your company can:

- **Lower Production Costs:** Utilize cost-effective microcontrollers with smaller memory capacities without sacrificing performance.
- **Extend Vehicle Range:** Higher precision (0.06% error) ensures the motor runs at its absolute loss-optimal state, reducing energy waste.
- **Scale for Complexity:** Easily incorporate additional inputs like temperature or magnetic flux variations that would be impossible with traditional tables.

FEAAM can assist your team by embedding this lightweight neural network into your existing microcontrollers, eliminating traditional memory bottlenecks to deliver superior motor efficiency and extended vehicle range without requiring expensive hardware upgrades.